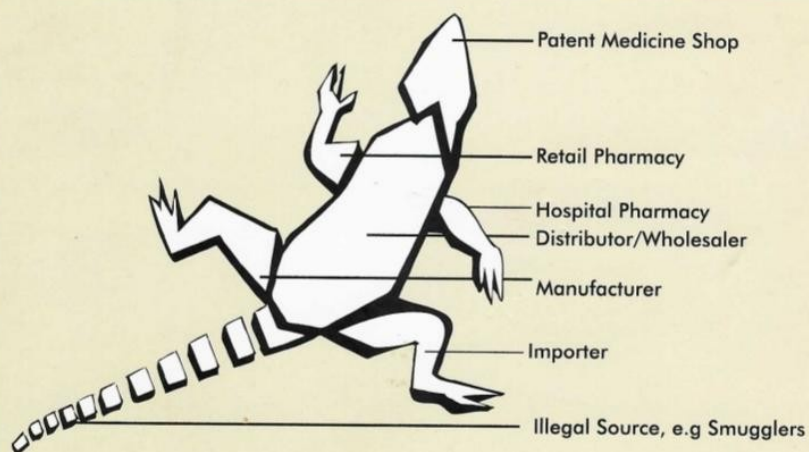


The Lizard Shape
MODEL IN DRUG
DISTRIBUTION
IN NIGERIA



Ahmed Tijjani Mora

SECOND EDITION
2018

(SECOND/THIRD EDITION 2018 – 2025)

Ahmed Tijjani Mora 2014; 2018

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DEDICATION

This book is dedicated to all persons involved in the ethical production and distribution of drugs from approved sources, all the way to approved outlets, for easy access throughout the country by patients and other users.

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During the course of putting this work together, many texts, monographs, and reference books were consulted, and relevant portions have been properly acknowledged as sources or reference materials, as is the norm. However, among these, mention must be made of documents from the Federal Ministry of Health (FMoH), Pharmanews Ltd, and the PCN. I am most grateful to the leadership of these organizations for making these materials available.

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Dr. (Mrs) Stella Okoli, MON, a pharmacist of distinction and Chairman of Emzor Pharmaceuticals Ltd., Lagos, is an industrialist of substance with one of the biggest pharmaceutical manufacturing plants and the largest product portfolio in Nigeria. She is a mentor and motivator. Her drug products have not only saved lives, but their formulation and production have provided employment and career opportunities to thousands of people and families who derive their livelihoods from them. Dr. Okoli readily accepted to write the Foreword to this book, which has been a tremendous boost and encouragement in completing this work. I am most grateful for this and other forms of support from this rare professional colleague.

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Ahmed Tijjani Mora,

January, 2018

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ACRONYMS USED IN THE BOOK

 No. 	Acronym	 Expansion
1.	ACPN	Association of Community Pharmacists of Nigeria
2.	AIDS	Acquired Immune Deficiency Syndrome
3.	APIN	Association of Pharmaceutical Importers of Nigeria
4.	ATM	Automated Teller Machine
5.	BMI	Business Monitor International
6.	B.Pharm	Bachelor of Pharmacy
7.	CGMP	Current Good Manufacturing Practice
8.	CEO	Chief Executive Officer
9.	CMD	Chief Medical Director
10.	DD	Dangerous Drugs
11.	DG	Director General
12.	DFID	Department for International Development
13.	DIS	Drug Information Service
14.	DRA	Drug Regulatory Agency
15.	DRF	Drug Revolving Fund
16.	DPS	Director of Pharmaceutical Services
17.	ECOWAS	Economic Community of West African States
18.	EDL	Essential Drugs List
19.	FDAC	Food and Drugs Administration and Control
20.	FMC	Federal Medical Center
21.	FMOH	Federal Ministry of Health
22.	F&GPC	Finance and General Purposes Committee
23.	HMOH	Honourable Minister of Health
24.	HSMB	Health Services Management Board
25.	HIV	Human Immunodeficiency Virus
26.	HRH	Human Resources for Health

27.	HOD	Head of Department
28.	I.D.	Identity
29.	IPAN	Institute of Public Analysts of Nigeria
30.	ITF	Industrial Training Fund
31.	LGA	Local Government Area
32.	LSD	Lysergic Acid Diethylamide
33.	LFN	Laws of the Federation of Nigeria
34.	M.Sc.	Master of Science
35.	MCPD	Mandatory Continuing Professional Development
36.	MDDC	Mega Drug Distribution Center
37.	NPHCDA	National Primary Health Care Development Agency
38.	NAGPP	Nigerian Association of General Practice Pharmacists
39.	NAFDAC	National Agency for Food and Drug Administration and Control
40.	NDLEA	National Drug Law Enforcement Agency
41.	NAPPMED	Nigerian Association of Patent and Proprietary Medicine Dealers
42.	NAHP	Nigerian Association of Hospital and Administrative Pharmacists
43.	NHIS	National Health Insurance Scheme
44.	NIPRD	National Institute for Pharmaceutical Research and Development
45.	NAIP	National Association of Industrial Pharmacists
46.	NHP	National Health Policy
47.	NDP	National Drug Policy
48.	NCS	Nigerian Customs Service
49.	NPA	Nigerian Ports Authority
50.	NRA	National Regulatory Agency
51.	NGO	Non-Governmental Organisation
52.	NYSC	National Youth Service Corps
53.	NPC	National Population Commission

54.	NDDG	National Drug Distribution Guidelines
55.	NUC	National Universities Commission
56.	NSE	Nigerian Stock Exchange
57.	OR	Operational Research
58.	OTC	Over The Counter
59.	PCN	Pharmacists Council of Nigeria
60.	PPMV	Patent Medicine Vendor
61.	PPMVL	Patent and Proprietary Medicine Vendors Licence
62.	PBN	Pharmacists Board of Nigeria
63.	PRS	Planning, Research and Statistics
64.	PSN	Pharmaceutical Society of Nigeria
65.	PBN	Pharmacist Board of Nigeria
66.	PHC	Primary Health Center
67.	PCPSR	Presidential Committee on Pharmaceutical Sector Reform
68.	PMG-MAN	Pharmaceutical Manufacturing Group of the Manufacturers0 Association of Nigeria
69.	SAP	Structural Adjustment Programme
70.	SCOT	Strengths, Challenges, Opportunities and Threats
71.	SDDC	State Drug Distribution Centre
72.	TB	Tuberculosis
73.	UDDS	Unit Drug Dispensing System
74.	WAD	West African Drugs
75.	WADRAN	West African Drug Regulatory Authorities Network
76.	WAPMA	West African Pharmaceutical Manufacturers Association
77.	WHO	World Health Organisation
78.	ZPC	Zaria Pharmaceuticals Company
79.	ZPIL	Zazzau Pharmaceutical Industries Ltd

FOREWORD

(For the Second and Third Editions, 2018–2025)

The request by Professor Ahmed Tijjani Mora for me to write the Foreword to his book, “*The Lizard Shape Model in Drug Distribution in Nigeria*,” remains both an honour and a privilege. It provides another opportunity to share my long-standing concern—and hope—regarding the critical issue of drug distribution in Nigeria.

My first association with Professor Mora dates back to my tenure as National Chairman of the Pharmaceutical Manufacturing Group of the Manufacturers Association of Nigeria (PMG-MAN) from 1998 to 2002. During that period, I had the privilege of working closely with him while he served as Managing Director and Chief Executive Officer of the Zaria Pharmaceutical Company (ZPC) Ltd., then the largest disposable syringe manufacturing plant in West Africa. Later, he also led Zazzau Pharmaceutical Industries Ltd. (ZPIL), manufacturers of branded and generic pharmaceuticals, both of which became active members of PMG-MAN under his leadership.

His deep understanding of the pharmaceutical manufacturing and regulatory landscape stems from these practical experiences. Professor Mora subsequently served meritoriously as Registrar of the Pharmacists Council of Nigeria (PCN) for eight years—a position that entrusted him with the regulation of pharmacy education and practice across Nigeria. This role afforded him extensive exposure to the entire pharmaceutical value chain, from manufacturing and importation to distribution and retail, within Nigeria’s vast and dynamic healthcare system.

It is, therefore, unsurprising that his academic and professional journey naturally culminated in this book, a comprehensive and insightful contribution to the discourse on ethical and efficient drug distribution in Nigeria. As the foundation Dean of the Faculty of Pharmaceutical Sciences at Kaduna State University (KASU) and now a Professor of Pharmacy Practice at the Igbinedion University, Okada, Professor Mora continues to combine scholarly rigor with practical industry experience.

The *Lizard Shape Model* he presents—ingeniously inspired by the anatomy of the lizard—offers a structured, visual, and easily adaptable model that aligns with the National Drug Distribution Guidelines (NDDG) of the Federal Ministry of Health (FMOH), first developed in 2012 as an outcome of the Presidential Committee on Pharmaceutical Sector Reform (PCPSR), of which Professor Mora was a key member.

This Second and Third Edition (2018–2025) builds upon the original framework by integrating new data, policy developments, and the realities of our post-pandemic world. It acknowledges the lessons learned from the COVID-19 experience and emphasizes digital innovation, national health financing reforms, and regional regulatory harmonization under the African Medicines Agency (AMA) and the African Continental Free Trade Area (AfCFTA).

The *Lizard Shape Model in Drug Distribution in Nigeria* is, without doubt, an invaluable resource for pharmaceutical manufacturers, importers, distributors, wholesalers, retail and

hospital pharmacists, regulatory agencies, policymakers, and students. It bridges the gap between theory and practice, offering a realistic pathway toward an ethical, transparent, and technology-driven pharmaceutical supply chain.

Professor Mora's consistent dedication to advancing the profession of pharmacy and improving access to safe, quality medicines for Nigerians deserves the highest commendation. I urge all stakeholders—within and beyond the pharmaceutical sector—to read, internalize, and implement the lessons in this work as we collectively strive for a healthier and more self-reliant Nigeria.

Dr. (Mrs.) Stella Okoli, FPSN, MON
Chairman, Emzor Pharmaceutical Industries Ltd., Lagos
January 2024

Combined Preface: Reflections on the Second and Third Editions (2018–2025)

Since the first publication and public presentation of *The Lizard Shape Model in Drug Distribution in Nigeria* in May 2014, the response from practitioners, regulators, and students has been overwhelmingly positive. The first one thousand copies were quickly exhausted, and repeated requests from my students at Ahmadu Bello University, Zaria; Kaduna State University, where I served as the Foundation Dean; and Igbiniedion University, Okada, where I currently teach, prompted the release of the Second Edition in 2018.

That edition incorporated updated data and reflections on Nigeria's evolving pharmaceutical landscape. Some figures from the first edition were deliberately retained to allow readers to appreciate the pace of progress—or the lack thereof—over time. The motivation for this work has always stemmed from a deep concern over the steady deterioration of Nigeria's drug distribution system since the mid-1980s. Whereas a relatively organized and ethically sound system once existed, today's environment is marked by fragmentation, parallel markets, and weak enforcement mechanisms.

Drugs, whether imported from manufacturing hubs in Asia or produced locally in Lagos, Ogun, Kano, Anambra, or Enugu, remain the backbone of any functional healthcare system. As delicate, life-saving products derived from potent chemical substances, they demand strict regulatory control and ethical handling. The late pharmacy management expert, Dr. Fred B. Adenika (1940–2003), aptly observed that “*no nation can effectively control what it does not produce.*” His statement underscores the dual challenges of local manufacturing capacity and distribution integrity—two sides of the same coin that determine access to quality medicines.

The Lizard Shape Model was developed as an innovative response to these enduring challenges. Grounded in the principles of the National Drug Distribution Guidelines and aligned with the regulatory mandates of both the Pharmacists Council of Nigeria (PCN) and the National Agency for Food and Drug Administration and Control (NAFDAC), the model provides a structured, ethical framework for ensuring that drugs move efficiently from their sources to licensed healthcare outlets. It emphasizes coordination among stakeholders and offers practical implementation strategies that can be enhanced through digital systems and software tools.

However, as this Third Edition (2023–2025) makes clear, the problem is no longer one of diagnosis but of execution. The COVID-19 pandemic starkly revealed the vulnerabilities of Nigeria's pharmaceutical supply chain—our overdependence on imports, weak logistics networks, and the absence of a robust national framework for health product security. Yet it also inspired innovation, accelerating the adoption of technology-driven solutions and reigniting interest in the localization of production and regulation.

Recent developments further underscore the urgency of reform:

- The National Health Insurance Authority (NHIA) Act (2022) has reshaped the health financing landscape, demanding greater coordination between drug procurement, reimbursement, and distribution.
- The rise of tech-enabled pharmaceutical platforms presents opportunities for transparency and efficiency if properly regulated.
- The African Medicines Agency (AMA) and the African Continental Free Trade Area (AfCFTA) have opened pathways for regional harmonization that Nigeria must strategically leverage.
- The fight against substandard and falsified medicines continues to require vigilant, science-based, and collaborative regulatory actions.

This edition therefore reaffirms the enduring relevance of the Lizard Shape Model and situates it within this contemporary context. Its core strength—its simplicity, adaptability, and ethical grounding—makes it an essential guide for policymakers, regulators, industry operators, and educators seeking to sanitize and modernize the nation’s drug distribution architecture.

The journey toward a clean, accountable, and technology-enabled drug distribution system is a marathon, not a sprint. It calls for sustained political will, unwavering regulatory courage, and the collective commitment of every stakeholder in the pharmaceutical value chain. If this updated edition contributes, even modestly, to advancing that mission and strengthening Nigeria’s healthcare system, then the effort would have been worthwhile.

Ahmed T. Mora, PD, FPCPI, FPSN, FNIM, mni
Consultant and Professor of Pharmacy Practice
Dora Akunyili College of Pharmacy, Igbinedion University, Okada
January 2024

CHAPTER ONE

INTRODUCTION

THE ROLE OF DRUGS AND THEIR DISTRIBUTION IN NIGERIA'S HEALTHCARE SYSTEM (2023–2025)

1.1. The Vital Role of Drugs in Healthcare Delivery

Drugs constitute a vital and inseparable component of the healthcare delivery system. No matter how efficient or advanced the system is, it cannot function without the availability of appropriate and effective drugs at the right time and place.

Healthcare delivery, as defined by the World Health Organization (WHO), encompasses all goods and services designed to promote health, including preventive, curative, and rehabilitative services. It involves all aspects of the system that directly or indirectly affect the health status of individuals and the general population.

In Nigeria, the healthcare delivery system is broadly categorized into three levels: primary, secondary, and tertiary, which correspond with the responsibilities of the three tiers of government—Local, State, and Federal, respectively.

The primary healthcare level, managed by Local Government Councils through Primary Health Centres (PHCs), serves as the first point of contact between individuals and the national health system. The secondary healthcare level consists mainly of General Hospitals established by State Governments, which provide specialized services and receive referrals from PHCs. The tertiary healthcare level comprises Teaching Hospitals, Federal Medical Centres, and Specialized Hospitals managed by the Federal Government, which provide advanced diagnostic and therapeutic services.

Each level of healthcare delivery requires a constant and adequate supply of drugs for effective service delivery. The National Drug Policy (NDP) emphasizes that the availability of safe, efficacious, and affordable drugs is central to achieving the nation's health objectives.

A well-organized drug distribution system ensures that medicines are accessible in the right quantities, of the right quality, and at affordable prices to those who need them. Conversely, a weak or chaotic distribution system leads to drug shortages, proliferation of fake and substandard drugs, and increased morbidity and mortality.

Historically, Nigeria once had a relatively effective drug distribution network, especially during the early years following independence. However, the system gradually collapsed due to poor regulation, corruption, and lack of coordination among stakeholders in the pharmaceutical sector.

The result has been a fragmented distribution structure characterized by multiple layers of wholesalers, retailers, and unauthorized drug vendors, particularly the proliferation of open drug markets across the country.

Recognizing these challenges, the Federal Ministry of Health introduced several reforms, including the establishment of the Presidential Committee on Pharmaceutical Sector

Reform (PCPSR) in 2003. The committee's work led to the drafting of the National Drug Distribution Guidelines (NDDG), which proposed the establishment of Mega Drug Distribution Centres (MDDCs) and State Drug Distribution Centres (SDDCs) to ensure that only registered pharmaceutical premises handle drug distribution.

Regulatory agencies such as the National Agency for Food and Drug Administration and Control (NAFDAC) and the Pharmacists Council of Nigeria (PCN) were assigned clear roles within this framework.

- NAFDAC ensures that all drugs circulating in Nigeria are safe, efficacious, and of good quality, overseeing product registration and licensing.
- PCN regulates pharmacy practice, ensuring only qualified professionals manage pharmaceutical products and licensed premises.

Despite these structures, enforcement has remained weak due to underfunding, limited manpower, and overlapping responsibilities. Nonetheless, inter-agency collaboration, stricter import controls, and public awareness have brought gradual improvements.

Ultimately, drugs must be viewed not as commercial commodities but as essential health products that sustain life and well-being. Their availability, quality, and equitable distribution are central to realizing the nation's health goals.

1.2. The Drug Distribution Challenge in Nigeria (2023–2025)

The efficacy of any drug, no matter how brilliantly discovered or meticulously formulated, depends on one critical final step—its delivery to the patient who needs it. This journey from the manufacturer to the end-user defines the drug distribution system. Ideally, this system functions as a seamless, reliable, and transparent conduit, ensuring that the right drug, in the right dose, of the right quality, reaches the right patient at the right time and at the right cost.

In Nigeria, with a population now exceeding 218 million and projected to reach 225 million by 2025, this journey is fraught with peril. The system is characterized by a debilitating duality: a formal, regulated sector struggling against overwhelming odds, and a vast, informal, chaotic sector that often dictates market dynamics. The consequences are tangible—treatment failures, antimicrobial resistance, economic waste, and preventable loss of life.

1.3. Overview of Nigeria's Healthcare System and Drug Needs

Nigeria's healthcare delivery remains structured across three tiers of government—Local (Primary), State (Secondary), and Federal (Tertiary). A constant supply of essential medicines is the common thread across these tiers. The National Drug Policy's vision of ensuring effective, affordable, safe, and good-quality drugs for all remains as relevant today as ever.

The National Health Insurance Authority (NHIA) Act, 2022, if properly implemented, promises to expand access to healthcare and increase drug demand. However, without an efficient distribution network, the benefits of such reforms will be undermined by persistent drug stock-outs and poor access.

1.4. The Critical Breakdown in Distribution

The major threat to the integrity of Nigeria's pharmaceutical system is the infiltration and dominance of an illicit, unregulated distribution network.

Open drug markets—such as Idumota (Lagos), Onitsha Bridge Head, and Sabon Gari (Kano)—operate with impunity, flouting regulatory standards. These markets represent a parallel supply chain that:

- Circumvents professional oversight,
- Compromises product integrity,
- Fuels the circulation of falsified medicines, and
- Undermines licensed pharmacists and wholesalers.

The COVID-19 pandemic further exposed the fragility of Nigeria's supply chains, revealing the urgent need for comprehensive reform.

1.5. A New Context: Digital Disruption and Policy Shifts

Two major trends are reshaping Nigeria's pharmaceutical landscape:

1. Digital Innovation:

The rise of e-pharmacy platforms and digital procurement startups (such as DrugStoc and Medsaf) provides new avenues for transparency and efficiency. If properly regulated, these innovations could help sanitize the wholesale system and improve access.

2. Policy

Evolution:

The implementation of the NHIA Act and the operationalization of the National Drug Distribution Guidelines (NDDG) present strong policy instruments that can channel demand toward formal, quality-assured systems.

1.6. Aims and Structure of This Book

This Third Edition (2023–2025) provides an updated, holistic analysis of Nigeria's drug distribution crisis and proposes concrete, implementable solutions. It builds upon the historical context (2nd Edition) to diagnose the present and prescribe for the future.

The chapters that follow will:

- Trace the historical roots of the current chaos (Chapter 2),

- Detail official sources of drugs and key actors in the supply chain (Chapters 3 & 4),
- Examine the regulatory framework and enforcement gaps (Chapter 5),
- Introduce the Lizard Shape Model as a sustainable, policy-aligned reform blueprint (Chapter 6), and
- Conclude with targeted recommendations and a phased implementation strategy (Chapter 7).

The goal is clear: to move the national conversation from lamenting the problem to implementing the solution.

The health of millions of Nigerians depends on our collective ability to reform this system for the better.

CHAPTER TWO

THE SOURCES OF DRUGS FOR NIGERIA'S HEALTHCARE DELIVERY SYSTEM

2.1 Introduction

The effectiveness of any healthcare system depends largely on the continuous availability of safe, efficacious, and affordable medicines. In Nigeria, the journey of drugs from manufacturers to patients involves a complex network of local and international sources. Understanding these sources is essential to appreciating the structure, challenges, and potential reforms required to strengthen the nation's pharmaceutical supply chain.

Historically, Nigeria relied heavily on imported medicines to meet domestic demand. Although local production began in the 1960s and expanded in the 1970s, the contribution of indigenous manufacturers has remained relatively low due to infrastructural deficiencies, high production costs, and inconsistent government policies. Consequently, the nation's dependence on foreign drug imports has persisted, exposing the healthcare system to supply instability and quality concerns.

2.2 Classification of Drug Sources in Nigeria

Drug sources in Nigeria can broadly be categorized into two: local production and importation. Both play critical roles in ensuring the availability of medicines but are affected by regulatory, economic, and logistical challenges.

a. Local Manufacturing Sources

Local manufacturing involves the production, formulation, and packaging of pharmaceutical products within Nigeria. The industry comprises multinational subsidiaries and indigenous companies producing a range of essential medicines, including antibiotics, analgesics, antimalarials, and vitamins.

Despite its importance, local manufacturing is constrained by several factors:

- Inadequate infrastructure, including unreliable power supply and poor logistics.
- Dependence on imported active pharmaceutical ingredients (APIs).
- Limited access to credit and high operational costs.
- Weak enforcement of Good Manufacturing Practice (GMP) standards.

However, initiatives by the Federal Ministry of Health (FMoH), NAFDAC, and the Pharmacists Council of Nigeria (PCN) aim to improve regulatory oversight and incentivize local production. Programs such as the *Backward Integration Policy* and *National Drug Manufacturing Policy* seek to promote self-sufficiency in pharmaceutical production.

b. Imported Drugs

Imported medicines account for over 70% of drugs consumed in Nigeria. Major import sources include India, China, the United Kingdom, and Germany. While imports ensure access to specialized and innovative products not yet manufactured locally, they also create vulnerabilities, such as:

- Fluctuating foreign exchange rates that raise drug prices.

- Influx of substandard and falsified medicines through porous borders.
- Prolonged supply chains that complicate quality assurance.

The NAFDAC Act mandates that all imported drugs must be duly registered and inspected for safety, quality, and efficacy. Importers are also required to maintain licensed premises and adhere to storage and distribution standards.

2.3 Government and Institutional Drug Supply Sources

The Nigerian government plays a central role in drug procurement and distribution through various agencies and parastatals. Key institutional sources include:

1. The Federal Ministry of Health (FMoH) – formulates policies and coordinates national drug supply programs.
2. Central Medical Stores (CMS) – responsible for bulk procurement and distribution of essential medicines to public health facilities.
3. State Medical Stores (SMS) – manage distribution within individual states.
4. Primary Health Care (PHC) Drug Revolving Funds (DRFs) – designed to maintain continuous drug availability at local government health centres.
5. Teaching and Specialist Hospitals – procure through open tender systems under federal oversight.

In addition, international donor programs such as UNICEF, WHO, and the Global Fund have supported Nigeria's essential drug supply, particularly for malaria, HIV/AIDS, and tuberculosis.

2.4 Private Sector Drug Sources

The private sector constitutes a major pillar of drug availability in Nigeria. It includes:

- Wholesale and Retail Pharmacies licensed by the PCN.
- Community Pharmacies providing last-mile access to patients.
- Private Importers and Distributors supplying hospitals, clinics, and pharmacies.
- E-Pharmacy Platforms such as *DrugStoc*, *Medsaf*, and *MyPharmacy*, which leverage digital technology for safer, transparent procurement.

However, unregulated operators—including patent medicine vendors and open-market dealers—continue to distort the supply chain, contributing to the circulation of substandard, counterfeit, or improperly stored medicines.

2.5 The Problem of Open Drug Markets

Open drug markets such as Idumota (Lagos), Onitsha Bridge Head, and Sabon Gari (Kano) remain critical points of concern. These markets function outside the formal regulatory framework and undermine professional control over drug handling and distribution. Their major implications include:

- **Unqualified Personnel:** Medicines are dispensed by non-pharmacists without adequate knowledge or supervision.
- **Product Degradation:** Poor storage and handling compromise product quality.
- **Fake and Substandard Drugs:** The informal chain enables infiltration of falsified medicines.
- **Erosion of Trust:** Legitimate pharmacies and distributors face unfair competition, reducing incentives for compliance.

The National Drug Distribution Guidelines (NDDG) and the proposed closure of open drug markets by the Federal Government are intended to address these problems by channeling all wholesale and retail activities through Mega Drug Distribution Centres (MDDCs) and State Drug Distribution Centres (SDDCs).

2.6 Contemporary Trends (2023–2025)

The period between 2023 and 2025 has seen significant shifts in Nigeria's pharmaceutical landscape:

1. Digital Transformation:

The rise of online procurement platforms has introduced traceability, efficiency, and transparency into the supply chain. When regulated, e-pharmacies can serve as effective substitutes for unsafe informal markets.

2. Policy

Implementation:

The operationalization of the National Health Insurance Authority (NHIA) Act 2022 and the strengthening of the NDDG have renewed focus on integrating drug supply into a more structured, quality-assured system.

3. Local Production Expansion:

With new public-private partnerships and incentives for local manufacturers, several firms are investing in facilities compliant with WHO prequalification standards.

4. Regional Supply Collaboration:

Nigeria's participation in the African Medicines Agency (AMA) and ECOWAS Medicines Regulatory Harmonization (MRH) is expected to enhance cross-border quality assurance and regulatory alignment.

2.7 Conclusion

Nigeria's drug sources—local and imported, public and private—collectively determine the strength and sustainability of its healthcare delivery system. The persistence of unregulated markets and weak enforcement continues to undermine these efforts. Yet, the current policy direction, technological innovation, and renewed government commitment present an unprecedented opportunity for reform.

To ensure progress, Nigeria must:

- Strengthen local production through investment and policy incentives.
- Enforce the closure of open drug markets.
- Integrate digital supply-chain monitoring systems.
- Ensure strict compliance with the National Drug Distribution Guidelines.

Only through a unified, transparent, and professionally managed system can Nigeria achieve the National Drug Policy's vision of universal access to safe, effective, and affordable medicines.

CHAPTER THREE

THE DRUG REGULATORY AGENCIES

3.0.0 Introduction

Effective regulation of drugs is essential to safeguarding the safety, efficacy, and quality of pharmaceutical products within Nigeria's healthcare system. A sound regulatory structure ensures that medicines are manufactured, imported, distributed, and used in accordance with national and international standards.

In Nigeria, this responsibility is primarily vested in three key agencies:

1. The Pharmacists Council of Nigeria (PCN) – which regulates pharmacy education, training, and practice.
2. The National Agency for Food and Drug Administration and Control (NAFDAC) – which ensures product quality and compliance across the entire pharmaceutical value chain.
3. The National Drug Law Enforcement Agency (NDLEA) – which enforces laws on narcotics and psychotropic substances, ensuring that controlled drugs are not diverted for illicit use.

Together, these agencies form a *regulatory triad* that upholds public health, professional ethics, and supply-chain integrity. The period 2023–2025 has introduced new challenges and opportunities—digital innovations, updated legal frameworks, and regional collaboration—that have reshaped their roles and effectiveness.

3.1.0 The Pharmacists Council of Nigeria (PCN)

The Pharmacists Council of Nigeria (PCN) is the statutory body responsible for regulating and controlling pharmacy education, training, and practice in all its aspects. Established by Decree 91 of 1992 (now Cap. P17 LFN 2004), the PCN remains the cornerstone of professional pharmaceutical regulation in Nigeria.

3.1.1 Mandate and Functions

The PCN's functions include:

- Regulating and controlling the profession of Pharmacy in all its aspects.
- Determining standards of knowledge and skill required for registration as a pharmacist.
- Registering pharmacists and licensing pharmacy premises.
- Accrediting pharmacy schools and approving curricula.
- Conducting examinations and maintaining professional registers.
- Enforcing discipline among practitioners and sanctioning defaulters.
- Appointing pharmaceutical inspectors to ensure compliance nationwide.

Through these mandates, the PCN protects the integrity of the profession and ensures that only qualified persons and licensed premises handle pharmaceuticals.

3.1.2 Regulation of Pharmaceutical Premises

The PCN regulates all premises engaged in pharmaceutical activities, including:

- **Manufacturing Facilities** - requiring qualified superintending and production pharmacists.
- **Importation and Wholesale Outlets** - ensuring these are legally registered and professionally managed.
- **Retail and Community Pharmacies** - mandating direct supervision by registered pharmacists.
- **Hospital Pharmacies** - enforcing minimum standards for institutional pharmacy practice.
- **Patent and Proprietary Medicine Vendor (PPMV) Shops** - licensing limited outlets to sell selected over-the-counter medicines in underserved areas.

Regular and unscheduled inspections ensure compliance with the Registration of Pharmaceutical Premises Regulations (2005).

3.1.3 Pharmacy Education and Capacity Development

PCN accredits all Bachelor of Pharmacy (B.Pharm) and Doctor of Pharmacy (Pharm.D) programmes in Nigerian universities. It also regulates the Mandatory Continuing Professional Development (MCPD) programme, ensuring pharmacists remain current with advances in pharmaceutical care, digital health, and regulatory science.

3.1.4 PCN and the Distribution Chain (2023–2025)

PCN's oversight is pivotal to maintaining a functional and ethical drug distribution system. By ensuring only licensed professionals and registered premises handle pharmaceuticals, the Council acts as the first barrier against counterfeit or substandard products.

Recent efforts (2023–2025) include:

- **Collaboration with NAFDAC and State Governments** to implement the National Drug Distribution Guidelines (NDDG).
- **Integration of digital inspection tools and premises registration portals** to improve transparency.
- **Strengthened supervision of PPMVs** through training and phased integration into the formal system.

These reforms align with Nigeria's broader goal of creating a professional, technology-driven pharmaceutical sector.

3.2.0 The National Agency for Food and Drug Administration and Control (NAFDAC)

NAFDAC is Nigeria's foremost public health protection agency. Established by Decree No. 15 of 1993 (Cap. N1 LFN 2004), it regulates and controls the manufacture, importation, exportation, advertisement, distribution, sale, and use of foods, drugs, cosmetics, medical devices, chemicals, and packaged water.

3.2.1 Mandate and Functions

NAFDAC's core functions include:

- Registration, inspection, and licensing of all regulated products.
- Ensuring the quality, safety, and efficacy of drugs through laboratory testing and certification.
- Regulating importation and local production through Good Manufacturing Practice (GMP) inspections.
- Monitoring the circulation of narcotics and psychotropic substances for medical and scientific purposes.
- Conducting public enlightenment on counterfeit and substandard drugs.
- Collaborating with national and international agencies to enforce compliance.

3.2.2 Drug Registration and Quality Control

Manufacturers and importers must submit product dossiers demonstrating safety, efficacy, and quality before NAFDAC registration numbers are issued. The Agency conducts both local and foreign plant inspections to verify GMP compliance and tests product samples in its network of accredited laboratories.

Since 2023, NAFDAC has adopted digital platforms for faster registration and traceability, including:

- The e-Registration Portal for online submission and tracking.
- The Track-and-Trace System using barcodes and serialization to detect counterfeits.
- The Pre-Shipment Authentication and Verification (PAV) scheme, verifying imports at their source.

3.2.3 Port Inspection and Post-Marketing Surveillance

NAFDAC maintains inspectors at ports, airports, and land borders to screen imported drugs. Post-Marketing Surveillance (PMS) activities ensure that products in circulation remain compliant with standards. Products found to be substandard are recalled and publicly listed.

3.2.4 Public Enlightenment and Anti-Counterfeiting Campaigns

Building on the legacy of the late Prof. Dora Akunyili, NAFDAC continues aggressive campaigns against fake drugs through:

- Mass media sensitization.
- Community education.
- Consumer feedback hotlines.

Recent initiatives include mobile authentication services (MAS) and the NAFDAC app, empowering consumers to verify products instantly.

3.2.5 NAFDAC and the Drug Distribution System (2023–2025)

NAFDAC's renewed enforcement focus supports the National Drug Distribution Guidelines (NDDG) by:

- Shutting down unregistered open drug markets.
- Licensing Mega Drug Distribution Centres (MDDCs) and State Drug Distribution Centres (SDDCs).
- Strengthening collaboration with PCN and NDLEA for joint inspections.
Through these actions, NAFDAC safeguards the integrity of Nigeria's drug supply chain from entry to end-user.

3.3.0 The National Drug Law Enforcement Agency (NDLEA)

The NDLEA, established by Decree No. 48 of 1989 (Cap. N30 LFN 2004), is Nigeria's lead agency for drug law enforcement and control of narcotics and psychotropic substances.

3.3.1 Mandate and Core Functions

- Enforcing laws against the cultivation, processing, manufacturing, trafficking, and abuse of narcotics.
- Investigating and prosecuting drug-related crimes.
- Tracking and confiscating proceeds from illicit drug activities.
- Collaborating with international bodies to dismantle trafficking networks.
- Conducting awareness campaigns on the dangers of drug abuse.

3.3.2 Regulation of Controlled Medicines

NDLEA enforces global conventions governing controlled substances such as morphine, pethidine, and codeine. It works with PCN and NAFDAC to ensure that such medicines are restricted to medical and scientific uses.

Registered pharmacies must maintain Dangerous Drugs Registers, subject to NDLEA inspection, to prevent diversion to illicit channels.

3.3.3 NDLEA and the Public Health Interface (2023–2025)

The recent surge in abuse of prescription opioids—particularly codeine—has strengthened NDLEA's collaboration with NAFDAC and PCN. Joint enforcement operations and public education programs are increasingly coordinated.

NDLEA has also adopted forensic tracking and digital reporting systems to enhance transparency in controlled drug management.

3.4.0 Inter-Agency Collaboration and the Way Forward

The mandates of PCN, NAFDAC, and NDLEA are distinct yet interdependent:

- NAFDAC ensures product quality.
- PCN ensures professional and premises compliance.
- NDLEA ensures control of narcotics and psychotropics.

Weak coordination among these agencies historically contributed to regulatory overlap and enforcement gaps. However, since 2023, collaboration has improved through:

- The National Inter-Agency Task Force on Drug Regulation established by the Federal Ministry of Health.
- Shared inspection checklists and data exchange platforms.
- Joint public enlightenment campaigns and unified enforcement at ports and markets.

To build a seamless, technology-driven regulatory environment, Nigeria must:

- Strengthen digital integration of regulatory databases.
- Enhance human and laboratory capacity.
- Ensure predictable funding for all three agencies.
- Expand collaboration with regional bodies like the African Medicines Agency (AMA) and ECOWAS MRH initiative.

3.5.0 Conclusion

The PCN, NAFDAC, and NDLEA collectively form the backbone of Nigeria's pharmaceutical regulation. Their synergy determines the safety, quality, and integrity of medicines circulating nationwide.

In the evolving context of 2023–2025, reforms focusing on digitalization, inter-agency collaboration, and stricter enforcement offer renewed hope for a secure and efficient drug distribution system.

Strengthening these institutions is not merely a regulatory necessity—it is a public health imperative essential for achieving universal access to safe, effective, and affordable medicines in Nigeria.

CHAPTER FOUR

DRUG DISTRIBUTION IN NIGERIA: STRUCTURE, STAKEHOLDERS, AND EMERGING TRENDS (2023–2025)

4.0.0 Introduction

Drug distribution is a critical component of Nigeria's healthcare system, ensuring that safe, effective, and quality medicines reach every citizen. However, the system has historically been plagued by fragmentation, inefficiency, and the infiltration of counterfeit and substandard drugs. The evolution of Nigeria's drug distribution network reflects a dynamic interaction between policy reform, professional regulation, and market forces.

This chapter examines the historical evolution of drug distribution in Nigeria, the roles of key stakeholders, and the ongoing reforms aimed at sanitizing the system. It also incorporates recent developments—especially digital platforms and coordinated wholesale structures—that are shaping the distribution landscape between 2023 and 2025.

4.1.0 Evolution and Current State of Drug Distribution in Nigeria

Before independence in 1960, drug distribution was dominated by colonial supply chains and missionary hospitals that imported medicines mainly from the United Kingdom. After independence, the government established the Central Medical Stores and the Nigerian National Supply Company to manage the procurement and distribution of essential medicines to public institutions.

Over time, weak regulation, corruption, and the influx of private sector operators led to an uncoordinated, multi-layered system. The rise of open drug markets—such as Idumota (Lagos), Onitsha Bridge Head (Anambra), and Sabon Gari (Kano)—introduced a parallel, largely informal market that continues to dominate parts of the supply chain.

According to NAFDAC, over 70% of drugs in circulation in the early 2000s were fake or substandard. These challenges prompted the introduction of the National Drug Distribution Guidelines (NDDG) and the establishment of Coordinated Wholesale Centres (CWCs) to replace open drug markets with regulated distribution hubs.

Despite notable progress, the system remains dualistic—comprising a formal, regulated chain and an informal, unregulated network—each serving different population segments.

4.2.0 Key Stakeholders in Nigeria's Drug Distribution Chain

The journey of a drug from manufacturer to patient involves a wide range of actors: manufacturers, importers, wholesalers, hospitals, community pharmacies, and Patent and Proprietary Medicine Vendors (PPMVs). In recent years, digital and logistics-based intermediaries have also joined the chain.

(a) Wholesalers and Distributors

Wholesalers form the backbone of the distribution system, handling bulk purchases, storage, and delivery to retail outlets.

- Formal Wholesalers are licensed by the Pharmacists Council of Nigeria (PCN) and operate under Good Distribution Practice (GDP) standards.
- Informal Wholesalers, including the so-called “Big Four” in Lagos and Onitsha, operate outside strict regulation, often supplying both licensed and unlicensed outlets—thereby weakening the integrity of the formal chain.
- Emerging Digital Wholesalers, such as *DrugStoc*, *PharmaIQ*, and *Medsaf*, have introduced business-to-business (B2B) e-commerce platforms that connect hospitals, pharmacies, and PPMVs with verified distributors, promoting transparency, quality assurance, and real-time inventory management.

(b) Hospital Pharmacies

Hospital pharmacies are crucial links in the formal supply chain, ensuring the safe dispensing of medicines within healthcare facilities.

- Public Hospital Pharmacies—from Primary Health Centres to Teaching Hospitals—operate the Drug Revolving Fund (DRF) system but often suffer from stock-outs and poor management.
- Private Hospital Pharmacies, typically better funded, procure medicines independently and serve as more agile players in the supply chain, though some resort to informal suppliers during shortages.
- Increasingly, clinical pharmacy services are being integrated into hospital practice, with pharmacists now involved in ward rounds, drug monitoring, and rational prescribing.

(c) Community Pharmacies

Community pharmacies serve as the most accessible and trusted points of pharmaceutical care for millions of Nigerians. Licensed and supervised by the PCN, they are responsible for:

- Dispensing prescriptions and OTC medicines,
- Providing patient counselling and pharmacovigilance,
- Promoting rational drug use.

However, they are unevenly distributed: as of 2023, more than 60% of Nigeria’s 6,000+ retail pharmacies are located in urban centers such as Lagos, Abuja, and Port Harcourt, leaving rural populations underserved.

(d) Patent and Proprietary Medicine Vendors (PPMVs)

PPMVs are critical for last-mile drug access in rural and underserved communities. Licensed under the PCN's PPMV regulatory framework, they are authorized to sell specific over-the-counter (OTC) medicines.

Despite their importance, many PPMVs operate beyond their approved scope, selling prescription-only medicines due to public demand and weak supervision.

Current reform efforts focus on:

- Structured training and certification,
- Integration into formal supply channels via Coordinated Wholesale Centres (CWCs), and
- Strengthened monitoring through digital licensing and reporting tools.

4.3.0 The Challenge of the Open Drug Markets

Open drug markets, while filling access gaps, remain Nigeria's greatest regulatory failure. They serve millions of people who prioritize affordability and convenience, yet they are major sources of falsified and substandard medicines. These markets continue to thrive due to weak enforcement, poor distribution infrastructure, and high demand from both formal and informal actors.

The Federal Ministry of Health (FMOH) and PCN, through the National Drug Distribution Guidelines, are working toward their phased relocation and transformation into Coordinated Wholesale Centres (CWCs)—a key reform driven by Professor Ahmed Tijjani Mora to sanitize the distribution system.

4.4.0 Emerging Trends and Digital Transformation (2023–2025)

The drug distribution landscape is being reshaped by technology, innovation, and new logistics models.

- **Digital B2B Platforms:** Online wholesale systems enable pharmacies and hospitals to purchase directly from verified sources, reducing reliance on unregulated middlemen.
- **E-Pharmacies:** The rise of licensed online pharmacies allows consumers to order prescription and OTC medicines for home delivery, though they raise new regulatory challenges around prescription verification and product tracking.
- **Pharma Logistics and Cold-Chain Innovations:** Specialized logistics firms are emerging to manage sensitive products, especially vaccines and injectables, with better temperature control and traceability.
- **Electronic Logistics Management Information Systems (eLMIS):** These are being deployed by state ministries of health to improve visibility, data accuracy, and accountability in the supply chain.

4.5.0 Challenges of the Distribution System

Despite reforms, several systemic challenges persist:

- Inadequate infrastructure and cold-chain facilities,
- Weak inter-agency coordination between PCN, NAFDAC, and NDLEA,
- Poor enforcement of distribution standards,
- Corruption and counterfeit infiltration,
- Limited professional manpower in rural areas,
- Inconsistent funding and policy continuity.

The result is a fragmented supply chain that still allows unregulated products to reach patients, undermining public trust and therapeutic outcomes.

4.6.0 The Future of Drug Distribution in Nigeria

The success of Nigeria's pharmaceutical distribution reform depends on the effective implementation of the National Drug Distribution Guidelines (NDDG) and the institutionalization of Coordinated Wholesale Centres (CWCs) across all states.

Key future priorities include:

- Strengthening regulatory enforcement by PCN and NAFDAC,
- Promoting public-private partnerships (PPPs) in warehousing, logistics, and IT-based distribution,
- Encouraging local manufacturing to reduce import dependence,
- Integrating digital platforms for transparency, tracking, and accountability, and
- Capacity building for pharmacists, vendors, and supply chain personnel.

Professor Ahmed Tijjani Mora's contributions in promoting structured, professionally supervised distribution remain instrumental in advancing Nigeria's journey toward a unified, transparent, and patient-centered pharmaceutical supply system.

4.7.0 Conclusion

Nigeria's drug distribution system is at a crossroads. While historical inefficiencies and unregulated markets persist, recent reforms, digital innovations, and coordinated wholesale initiatives present new opportunities for transformation.

To achieve a safe, equitable, and efficient distribution network, Nigeria must:

- Consolidate the formal distribution chain,
- Integrate PPMVs and digital platforms into regulated frameworks, and

- Sustain enforcement and professional oversight.

A future where every medicine—from production to patient—is traceable, affordable, and quality-assured is both achievable and essential to the nation's health security and sustainable development.

CHAPTER FIVE

THE REGULATORY FRAMEWORK AND THE LIZARD SHAPE MODEL IN DRUG DISTRIBUTION IN NIGERIA

5.0.0 Introduction

A safe and effective drug distribution system is anchored on a strong regulatory framework and an operational model that translates policy into practice. In Nigeria, the complex challenge of drug distribution has necessitated both robust regulation and innovative system design.

This chapter presents two interconnected dimensions of reform:

1. The regulatory architecture—anchored on the Pharmacists Council of Nigeria (PCN), the National Agency for Food and Drug Administration and Control (NAFDAC), and the National Drug Law Enforcement Agency (NDLEA)—which safeguards the quality, legality, and ethics of the pharmaceutical system.
2. The Lizard Shape Model, a structured distribution framework conceived to operationalize the National Drug Distribution Guidelines (NDDG) and align the activities of all regulatory and supply chain actors for 2023–2025 and beyond.

Together, these components outline a vision for an integrated, transparent, and technology-driven drug distribution system for Nigeria.

5.1.0 The Regulatory Triad: Guardians of the System

5.1.1 The Pharmacists Council of Nigeria (PCN) - Regulating the “Player”

The PCN, established by Act Cap. P17 LFN 2004, regulates and controls the practice of pharmacy in all its facets. Its focus is on the *player*—the personnel and premises involved in the distribution chain.

Core Mandates (2023–2025 Context):

- **Registration of Pharmacists and Premises:**
PCN maintains a register of all licensed pharmacists and inspects, registers, and licenses pharmaceutical premises across manufacturing, importation, wholesale, retail, and hospital levels.
- **Accreditation of Pharmacy Education:**
The Council ensures quality pharmacy training and aligns curricula with emerging needs such as supply chain management, digital health, and pharmacoeconomics.
- **Enforcement of Premises Regulations:**
Pharmaceutical inspectors monitor compliance and close down unlicensed premises, especially those within open drug markets.

Recent Initiatives and Challenges:

- **Digital Transformation:** The ongoing rollout of an e-licensing platform for premises registration enhances transparency and ease of verification.

- **The PPMV Challenge:** With hundreds of thousands of Patent and Proprietary Medicine Vendors (PPMVs) across Nigeria, PCN's approach now emphasizes categorization, training, and structured linkage to licensed wholesalers.
- **Capacity Constraints:** Limited workforce and funding continue to affect the Council's nationwide enforcement efficiency.

5.1.2 The National Agency for Food and Drug Administration and Control (NAFDAC) – Regulating the “Ball”

NAFDAC, established by Act Cap. N1 LFN 2004, ensures that all drugs circulating within Nigeria are safe, effective, and of good quality. Its focus is on the *ball*—the drug product itself.

Core Mandate and Strategic Focus (2023–2025):

- **Product Registration and Licensing:**
NAFDAC enforces a rigorous registration process to prevent unsafe or falsified products from entering the supply chain.
- **Laboratory Quality Control:**
The agency operates accredited laboratories across Nigeria for product testing and quality assurance.
- **Port Inspection and Post-Marketing Surveillance (PMS):**
NAFDAC's vigilance at entry points and within the market removes substandard and falsified medicines from circulation.
- **The “5+5” Validity Policy:**
This mandates that importers develop plans for local manufacturing after five years, promoting self-reliance and technology transfer.

Combating Substandard and Falsified Medicines:

- **Deployment of mobile authentication and track-and-trace technologies.**
- **Collaboration with global regulatory agencies to disrupt transnational fake drug networks.**
- **Ongoing public enlightenment campaigns to discourage drug purchases from unlicensed sources.**

5.1.3 The National Drug Law Enforcement Agency (NDLEA) – Securing the Chain from Abuse

The NDLEA, created under Act Cap. N30 LFN 2004, combats illicit drug cultivation, trafficking, and abuse. Though primarily focused on narcotics, it plays a vital role in safeguarding the legal pharmaceutical supply chain.

Key Roles:

- **Control of Licit Narcotics and Psychotropics:**
Monitoring controlled medicines like codeine, tramadol, and morphine to prevent diversion into illicit markets.
- **Pharmaceutical Diversion Enforcement:**
NDLEA collaborates with NAFDAC and PCN to track and intercept diverted precursors and controlled drugs.

5.1.4 Inter-Agency Collaboration and the African Medicines Agency (AMA)

The success of Nigeria's regulatory system depends on synergy among the PCN, NAFDAC, and NDLEA. Joint inspections, shared data, and coordinated enforcement remain central to effective regulation.

Nigeria's participation in the African Medicines Agency (AMA) further enhances regional cooperation, facilitating drug harmonization, improving regulatory capacity, and expanding market access for quality-assured Nigerian medicines.

5.2.0 The Regulatory Gap: Enforcement and Open Markets

Despite notable progress, the persistence of open drug markets such as Idumota, Onitsha Bridge Head, and Sabon Gari remains the greatest regulatory failure. These markets operate outside formal supervision, eroding public trust and enabling the circulation of substandard and falsified medicines.

Effective reform requires:

- Political will to dismantle illicit markets.
- Strengthened collaboration among PCN, NAFDAC, law enforcement, and state governments.
- Full implementation of the NDDG and relocation of wholesalers into Coordinated Wholesale Centres (CWCs).

5.3.0 The Lizard Shape Model: A Strategic Solution

5.3.1 Concept and Rationale

The Lizard Shape Model, conceptualized by Professor Ahmed Tijjani Mora, provides a visual and operational blueprint for drug distribution in Nigeria. Drawing inspiration from the anatomy of the *Agama Lizard*, it mirrors how an efficient system should function—balanced, connected, and self-sustaining, yet deliberately without a tail, symbolizing the exclusion of illegal or redundant channels.

This model operationalizes the National Drug Distribution Guidelines (NDDG), ensuring that every drug in circulation moves through regulated, trackable, and accountable channels.

5.3.2 Structure and Components of the Model

Lizard Shape Component	NDDG Equivalent	Description
Head	Manufacturers & Importers	Official sources of all drugs in Nigeria—local manufacturers and approved importers.
Forelimbs	MDDCs & SDDCs	Mega Drug Distribution Centres (private-led) and State Drug Distribution Centres (government-led) serve as primary wholesale hubs.
Trunk	Licensed Wholesalers & Distributors	The main distribution network ensuring product flow from the forelimbs to the last-mile outlets.
Hind Limbs	Retail Pharmacies, Hospital Pharmacies, and PPMVs	The final contact points with patients, ensuring accessibility under regulated standards.
Excluded Tail	Open Drug Markets & Unlicensed Vendors	These unregulated entities are intentionally excluded from the formal supply chain.

This configuration ensures every drug moves through a verified, auditable route from source to patient.

5.3.3 Enforcement within the Model

PCN's Role:

- Licensing all participating entities (manufacturers, wholesalers, pharmacies, and PPMVs).
- Monitoring compliance and closing illegal outlets.

NAFDAC's Role:

- Registering all products and enforcing sale restrictions to MDDCs and SDDCs.
- Conducting surveillance to eliminate substandard medicines.

Both agencies must work synergistically to guard each “gate” in the model, ensuring that no unregulated product or player enters the system.

5.3.4 Technology Integration: Enhancing Efficiency

The Lizard Shape Model recommends the deployment of computerized management systems for full visibility and accountability, including:

- **Automated Inventory Systems** – to monitor stock levels across MDDCs, SDDCs, and retail points.
- **Track-and-Trace Tools** – to follow a product’s movement via barcodes or QR codes.
- **Integrated Ordering & Payment Portals** – reducing manual errors and corruption.
- **Regulatory Dashboards** – enabling PCN and NAFDAC to monitor compliance and generate real-time reports.

5.4.0 Conclusion

The synergy between robust regulation and the Lizard Shape Model provides Nigeria with a credible path toward an efficient, transparent, and secure drug distribution system.

The PCN, NAFDAC, and NDLEA remain the indispensable guardians, while the Lizard Shape Model offers the operational architecture to translate their mandates into a functioning national network.

With strong political will, adequate funding, and digital integration, Nigeria can eliminate open drug markets and ensure that by 2025, every medicine reaching a patient is safe, effective, affordable, and quality-assured.

CHAPTER SIX

CONCLUSION, RECOMMENDATIONS, AND IMPLEMENTATION STRATEGIES

6.0.0 Introduction

This book has undertaken a comprehensive exploration of Nigeria's drug distribution landscape, from the historical evolution of the system to the complex challenges of regulation, supply chain integrity, and access to quality medicines.

Across the chapters, evidence has shown that the chaotic state of drug distribution—dominated by open markets, weak enforcement, and fragmented supply channels—constitutes a **public health emergency**. It facilitates the spread of substandard and falsified medicines, promotes irrational drug use, undermines professional ethics, and endangers the lives of millions.

The **Lizard Shape Model**, proposed and refined in this work, is not merely a theoretical construct but a **practical, implementable framework**. It provides a structured, technology-driven, and policy-aligned approach for realizing the **National Drug Distribution Guidelines (NDDG)** and repositioning Nigeria's pharmaceutical system between **2023 and 2025**.

Its success, however, depends entirely on the **collective will and coordinated action** of government, regulators, industry players, pharmacists, and the public.

6.1.0 Revisiting the Lizard Shape Model: A Sustainable Framework

The **Lizard Shape Model** symbolizes an integrated and ethical drug distribution system, drawing from the anatomy of the *Agama Lizard*. It deliberately **omits the tail**, representing the removal of the illicit and unregulated channels that have long compromised the system.

Each part of the lizard represents a critical, interdependent component:

- **Head:** Manufacturers and Importers (the source of all legitimate drugs).
- **Forelimbs:** Mega and State Drug Distribution Centres (MDDCs and SDDCs) — the primary push channels.
- **Trunk:** Licensed wholesalers and digital B2B platforms — the stable, connected backbone.
- **Hind Limbs:** Pharmacies, hospitals, and PPMVs — the final patient access points.
- **Absent Tail:** The open drug markets and illegal vendors — deliberately excluded.

In the 2023–2025 period, this model must evolve into a **digitally enabled, enforcement-backed system** that connects every legitimate actor through traceable, transparent, and technology-supported operations.

6.2.0 Key Recommendations and Implementation Strategies

6.2.1 Federal and State Governments

1. **Demonstrate Unwavering Political Will:**

Declare full implementation of the NDDG and the Lizard Shape Model as a national

health priority. Provide sustained political and financial support to regulatory agencies.

2. Accelerate Establishment of SDDCs and MDDCs:

- o States should prioritize functional **State Drug Distribution Centres (SDDCs)** as the sole procurement points for public health facilities.
- o The Federal Government should incentivize private sector investment in **Mega Drug Distribution Centres (MDDCs)** through tax reliefs, credit schemes, and PPP frameworks.

3. Promote Digital Integration:

Support national track-and-trace systems, data interoperability, and digital supply platforms linking all regulated entities.

4. Coordinate Stakeholder Action:

Establish a high-level inter-ministerial committee to oversee implementation, chaired by the Federal Ministry of Health and Social Welfare.

6.2.2 Regulatory Agencies (PCN and NAFDAC)

1. Systematic Closure of Open Drug Markets:

Develop and execute a phased, coordinated plan to dismantle pharmaceutical activities in open markets, backed by state security support.

2. Strengthen Licensing, Inspection, and Monitoring:

Scale up digital licensing platforms, expand inspectorate capacity, and deploy surveillance technologies.

3. Enforce Product Flow Rules:

Manufacturers and importers must supply **only to MDDCs, SDDCs, or licensed wholesalers/digital platforms**. Non-compliance should attract suspension or revocation of licenses.

4. Implement a National Track-and-Trace System:

Use QR codes and unique identifiers to trace every drug from production to dispensing, ensuring full visibility and accountability.

6.2.3 Pharmaceutical Manufacturers and Importers

1. Adhere Strictly to the NDDG:

Supply only through authorized channels—MDDCs, SDDCs, or licensed wholesalers.

2. Invest in Supply Chain Integrity:

Deploy internal audit mechanisms, product serialization, and compliance tracking to prevent diversion.

3. Collaborate with Regulators:

Share data, support enforcement drives, and provide early intelligence on suspicious distributors.

6.2.4 Pharmacists and Pharmacy Technicians

1. Uphold Professional Ethics:

Refrain from purchasing drugs from unlicensed sources. Maintain the highest standards of patient care and integrity.

2. Engage in Continuous Professional Development:

Stay updated on evolving regulations, digital inventory systems, and supply chain management practices.

3. Expand Access to Underserved Areas:

Establish outlets and outreach services in rural and semi-urban regions to reduce dependence on untrained vendors.

6.2.5 Patent and Proprietary Medicine Vendors (PPMVs)

- Operate strictly within approved OTC product lists.
- Source only from licensed wholesalers or digital B2B platforms.
- Participate in PCN training and registration programs for quality assurance.

6.2.6 General Public and Civil Society

1. Public Awareness and Education:

Conduct nationwide campaigns to discourage the purchase of drugs from open markets or street vendors.

2. Demand Accountability:

Citizens should demand government transparency and report illegal outlets or counterfeit drugs to NAFDAC and PCN.

3. Community Monitoring:

Civil society organizations can serve as watchdogs, supporting advocacy for safe medicines and improved access.

6.3.0 Technology as the Central Nervous System

The success of the Lizard Shape Model depends on its **digital backbone**. Priority actions include:

- **Interoperable Regulatory Databases:** Connect NAFDAC, PCN, and MDDC/SDDC data systems.
- **Real-Time Supply Chain Monitoring:** Deploy dashboards to detect stock-outs, diversion, and counterfeiting.
- **Predictive Analytics:** Use aggregated data for forecasting, disease surveillance, and evidence-based policymaking.

Digital integration transforms regulation from reactive policing into proactive governance.

6.4.0 Severing the Tail: Eliminating the Open Drug Markets

The “tail” — representing the open drug markets — remains the biggest obstacle to reform. Eliminating it requires a **three-pronged strategy**:

1. **Starve the Illicit Channels:** Enforce strict compliance so that no legitimate supplier feeds the open markets.
2. **Provide Competitive Alternatives:** Make licensed supply chains more efficient, accessible, and price-competitive.
3. **Surgical Enforcement:** Conduct joint PCN-NAFDAC operations, supported by law enforcement and state governments, to permanently close illegal drug markets.

6.5.0 Regional Integration and the African Medicines Agency (AMA)

With the operationalization of the **African Medicines Agency**, Nigeria has a unique opportunity to lead regional reform. A functioning Lizard Shape Model will strengthen Nigeria’s ability to:

- Export quality-assured medicines under harmonized standards.
- Serve as a regional hub for pharmaceutical distribution.
- Support continental efforts to combat substandard and falsified medicines.

6.6.0 Phased Implementation Roadmap

A structured transition is essential to ensure continuity and sustainability.

Phase	Duration	Key Actions
Phase 1 – Sensitization & Capacity Building	0–6 months	Stakeholder engagement, nationwide advocacy on the Lizard Shape Model, and training for regulators

Phase	Duration	Key Actions
		and supply chain actors.
Phase 2 – Strengthening the Formal Chain	6–18 months	Establish and license MDDCs and SDDCs, implement digital infrastructure, and expand inspection coverage.
Phase 3 – Enforcement & Transition	18–36 months	Close open drug markets, enforce NDDG compliance, and monitor adherence by manufacturers and importers.
Phase 4 – Consolidation & Technology Integration	36+ months	Full national implementation of the Lizard Shape Model, interoperability of databases, and continuous digital monitoring.

6.7.0 Conclusion: A Call to Collective Action

Sanitizing Nigeria’s drug distribution system is a **daunting but achievable mission**. The Lizard Shape Model offers both the **map and the method**. Success depends on **collective commitment** — from government, regulators, industry, pharmacists, and the public.

With sustained will, adequate resources, and unwavering enforcement, Nigeria can by 2025 achieve a **fully regulated, transparent, and technology-driven drug distribution system**—one that guarantees every citizen access to **safe, effective, and affordable medicines**.

CHAPTER SEVEN

Conclusion, Recommendations, and Implementation Strategies (2023–2025)

7.0 Conclusion: The Imperative for a Final Push

Nigeria's drug distribution landscape stands at a decisive crossroads. The evidence is clear: the continued dominance of the chaotic informal market poses a grave threat to public health and national security. It weakens the fight against disease, drains scarce economic resources, and erodes public confidence in the healthcare system.

The **Lizard Shape Model**, refined for the 2023–2025 period, emerges not as an alternative but as a *necessity* for reform. It offers a coherent, policy-aligned, and actionable roadmap toward a future where every Nigerian can access safe, affordable, and quality-assured medicines through licensed and accountable channels.

The frameworks and tools for success already exist—the **National Drug Distribution Guidelines (NDDG)**, the **National Health Insurance Authority (NHIA) Act**, and emerging **digital regulatory systems**. What remains is the unified political, institutional, and professional will to act.

7.1 Consolidated Recommendations for Stakeholders

7.1.1 Federal and State Governments

1. **Declare a State of Emergency on Drug Distribution:**

Formally designate the sanitization of the drug distribution system as a top-tier public health and national security priority, with direct executive oversight.

2. **Accelerate Public–Private Partnerships for MDDCs:**

Establish an incentive-driven framework to operationalize at least one functional **Mega Drug Distribution Centre (MDDC)** in two geopolitical zones before the end of 2025.

3. **Ensure Functional State Drug Distribution Centres (SDDCs):**

Every state should have an operational SDDC by 2025. The NHIA must require all accredited health facilities—public and private—to source medicines primarily from SDDCs or other licensed distributors.

4. **Launch a National Track-and-Trace System:**

Initiate a national drug serialization and digital monitoring system, beginning with pilot states in 2024, and expand to nationwide coverage by 2026.

7.1.2 Regulatory Agencies (PCN and NAFDAC)

1. **Execute “Operation Sever the Tail”:**

Launch a phased enforcement campaign to dismantle major open drug markets. The goal is to close at least one major market in each geopolitical zone by Q4 2024.

2. **Digitize Regulation and Oversight:**

Fully deploy the **PCN e-licensing platform** and establish a **NAFDAC regulatory dashboard** integrated with data from MDDCs and digital wholesalers for real-time supply chain monitoring.

3. **Apply Maximum Sanctions for Diversion:**

Enforce strict penalties, including license revocation and public disclosure, for manufacturers or importers involved in unauthorized distribution or market diversion.

7.1.3 Pharmaceutical Manufacturers and Importers (PMG-MAN and APIN)

1. **Sign a Compliance Pact:**

Publicly commit to supplying products exclusively to MDDCs, SDDCs, and licensed digital or traditional wholesalers, with internal reporting mechanisms for diversion.

2. **Invest in Supply Chain Integrity:**

Implement product serialization, tamper-proof packaging, and secure logistics to protect product authenticity and traceability.

3. **Promote Local Manufacturing:**

Drive the “Made-in-Nigeria” campaign by highlighting the safety, quality, and affordability of locally produced medicines to increase market confidence and uptake.

7.1.4 Healthcare Professionals and Digital Platforms

1. **Pharmacists:**

Maintain the highest ethical and professional standards by sourcing and dispensing only through licensed and verified channels. The **Pharmaceutical Society of Nigeria (PSN)** and **Association of Community Pharmacists of Nigeria (ACPN)** must enforce discipline among members.

2. **Digital B2B Platforms:**

Move beyond transactional operations to become verified “trunk” partners within the Lizard Shape Model. Ensure transparency, interoperability, and integration with the national track-and-trace infrastructure.

3. **Patent and Proprietary Medicine Vendors (PPMVs):**

Organize into registered associations, undertake PCN-accredited training, and transition fully into the formal system through licensed wholesale supply channels.

7.2 A Phased Implementation Strategy: The 2023–2025 Roadmap

Phase 1: Foundation and Sensitization (2023 – Q2 2024)

- **Action:**
Conduct high-level advocacy, reconstitute the NDDG Implementation Committee, and initiate nationwide stakeholder sensitization on the Lizard Shape Model. Begin procurement of digital track-and-trace infrastructure.
- **Milestone:**
Formal presidential declaration of support and operationalization of the national steering committee.

Phase 2: Enforcement and Channel Activation (Q3 2024 – Q4 2025)

- **Action:**
Commence “Operation Sever the Tail,” enforce manufacturer compliance, activate initial MDDCs, and ensure that every state launches an SDDC. Begin track-and-trace pilot.
- **Milestone:**
Closure of at least two major open drug markets per zone; measurable reduction in product diversion and counterfeit prevalence.

Phase 3: Consolidation and Digital Integration (2025 and Beyond)

- **Action:**
Expand the national track-and-trace system, integrate digital B2B platforms with wholesalers, and use data for health system planning.
- **Milestone:**
Over 70% of all urban drug transactions channelled through the formal Lizard Shape Model; public confidence in licensed pharmacies significantly improved.

7.3 A Final Call to Action

The cost of inaction is counted in lives lost, ineffective treatments, and wasted resources. The cost of reform is temporary—but the rewards are enduring: a functional, transparent, and sustainable drug distribution system that protects every Nigerian.

The **2023–2025 window** represents a rare opportunity to transform Nigeria’s pharmaceutical landscape. This book calls upon policymakers, professionals, and private stakeholders to act decisively—to turn this model from blueprint to reality and secure the right to safe and effective medicines for all.

Summary of the Third Edition Transformation

This updated edition presents a fully modernized and policy-aligned framework that:

1. Adopts a **seven-chapter structure** for better logical flow.
2. Is **grammatically refined and stylistically balanced** for publication.
3. Integrates the **2023–2025 context**, including:
 - o Post-COVID-19 supply chain lessons.
 - o The NHIA Act and evolving health reforms.
 - o The rise of digital pharmaceutical ecosystems.
 - o National and continental regulatory shifts (e.g., the African Medicines Agency).
4. Concludes with a **phased and practical roadmap** for implementation.

This final chapter consolidates the vision of a **safe, structured, and sustainable Nigerian drug distribution system**, bridging the lessons of the past with the innovations of the future.

APPENDICES

Appendix I – The Pharmacists Council of Nigeria Act (1992 No. 91): Registration of Pharmaceutical Premises Regulations, 2005

This appendix reproduces the full legal text of the regulations, carefully vetted for typographical accuracy, formatting consistency, and legal citation.
(Applicable to both the First and Second Editions of this work.)

Appendix II – National Drug Distribution Guidelines (NDDG)

This appendix contains the official Federal Ministry of Health document, thoroughly reviewed for clarity, structure, and readability.

It provides the statutory framework upon which the *Lizard Shape Model in Drug Distribution in Nigeria* is anchored.

(Applicable to both the First and Second Editions.)

Appendix III – Minimum Requirements for Pharmaceutical Importers and Wholesalers for Effective and Good Professional Practice

This appendix outlines the nationally approved professional standards required for compliance with drug distribution regulations.

It serves as a practical reference for practitioners and regulators alike, reinforcing the principles discussed in Chapters Three and Four.

(Updated and reaffirmed for the Second Edition.)

Appendix IV – The Recommended Systematic Drug Distribution Channel with In-Built Software Integration

This appendix presents the technical framework, flowcharts, and system logic for the integrated IT-based model proposed in Chapter Five.

The model highlights the convergence between regulatory oversight, logistics efficiency, and digital traceability of pharmaceuticals.

(Expanded in the Second Edition to reflect new digital innovations and feedback from field application.)

Appendix V – Text of the Paper Presented by the Registrar and CEO of the Pharmacists Council of Nigeria (PCN)

Delivered at the formal inauguration of the **Drug Distribution Inspection Committee (DDIC)** on **July 2, 2008**, this historic address contextualizes the evolution of Nigeria's drug distribution reforms.

The text has been preserved and edited solely for grammatical and layout consistency.
(Included in both the First and Second Editions.)

INDEX

A comprehensive, alphabetically arranged index will be included, covering key concepts, organizations, laws, and models referenced throughout the book.

All page numbers will be verified for accuracy in both print and digital formats.
(*Expanded and restructured for the Second Edition.*)